SETUP

The project is modelled as a Work Breakdown Structure (WBS)-organised directed acyclic graph of n activities with finish-to-start (FS), zero-lag precedence relationships. For each activity i, $P(i)$ denotes its set of immediate predecessors and $d_i > 0$ its baseline duration in working days. The methodology follows AACE International Recommended Practice 57R-09 (*Integrated Cost and Schedule Risk Analysis*) and replicates the core iteration engine of Oracle Primavera Risk Analysis (OPRA).

DETERMINISTIC BASELINE (CPM FORWARD PASS)

Given a topological ordering of the activity network, the Critical Path Method (CPM) forward pass computes the early start (ES_i) and early finish (EF_i) of each activity recursively:

$$ES_i = \begin{cases} 0 & P(i) = \emptyset \\ \max\limits_{j \in P(i)} EF_j & \text{otherwise} \end{cases}, \qquad EF_i = ES_i + d_i, \qquad T^{\text{det}} = \max_i EF_i. \tag{1}$$

T^{det} is the deterministic project finish under the assumption that every activity executes at exactly its baseline duration.

DURATION UNCERTAINTY MODEL

Each activity receives a *three-point estimate* consisting of multiplicative optimistic, most-likely, and pessimistic factors $(\alpha_i, \mu_i, \beta_i)$ with $0 < \alpha_i \le \mu_i \le \beta_i$, yielding

$$(a_i, m_i, b_i) = (\alpha_i d_i,\ \mu_i d_i,\ \beta_i d_i). \tag{2}$$

Activity duration is modelled as a triangular random variable $D_i \sim \mathrm{Tri}(a_i, m_i, b_i)$ with density

$$f_{D_i}(x) = \begin{cases} \dfrac{2(x - a_i)}{(b_i - a_i)(m_i - a_i)} & a_i \le x \le m_i, \\ \dfrac{2(b_i - x)}{(b_i - a_i)(b_i - m_i)} & m_i < x \le b_i, \\ 0 & \text{otherwise.} \end{cases} \tag{3}$$

Risk-register findings are encoded into distribution *width*: activities exposed to multiple risks receive wider pessimistic tails (β_i). Consistent with AACE RP 57R-09, *discrete risk events* (explicit probability–impact pairs) are not modelled; all risk is represented through duration uncertainty, trading event-level fidelity for parameter transparency and auditability.

MONTE CARLO PROCEDURE

For N iterations (default $N = 10{,}000$, fixed pseudo-random seed for reproducibility), each iteration k samples independent durations $d_i^{(k)} \sim \mathrm{Tri}(a_i, m_i, b_i)$ for every activity, executes the CPM forward pass to obtain $EF_i^{(k)}$, and records the project finish $T^{(k)} = \max_i EF_i^{(k)}$.

OUTPUT STATISTICS AND RISK EXPOSURE

The empirical distribution $\{T^{(k)}\}_{k=1}^N$ yields

$$\bar{T} = \frac{1}{N}\sum_{k=1}^{N} T^{(k)}, \quad s_T^2 = \frac{1}{N}\sum_{k=1}^{N}(T^{(k)} - \bar{T})^2, \quad \hat{F}_T(t) = \frac{1}{N}\sum_{k=1}^{N} \mathbf{1}\{T^{(k)} \le t\}. \tag{4}$$

Confidence levels are reported as percentiles $P_p = \hat{F}_T^{-1}(p/100)$ for $p \in \{5, 10, \ldots, 95\}$ and visualised on the histogram and the S-curve (empirical CDF). Two scalars carry primary operational weight: the **deterministic confidence** $\hat{F}_T(T^{\text{det}})$ — a low value ($\ll 50\%$) indicates the baseline is unreachable — and the **risk exposure at P80**, $\Delta_{80} = P_{80} - T^{\text{det}}$, which represents the cumulative schedule buffer required to achieve 80% delivery confidence and serves as the basis for commitment-date negotiation.

ACTIVITY SENSITIVITY (TORNADO)

For each activity i, the driver effect on project finish is quantified by the Spearman rank correlation between its sampled duration and the iteration finish:

$$\rho_i = 1 - \frac{6\sum_{k=1}^{N}\big(R(d_i^{(k)}) - R(T^{(k)})\big)^2}{N(N^2 - 1)}, \tag{5}$$

where $R(\cdot)$ denotes the rank within the N-sample. Spearman is preferred over Pearson for robustness to monotone nonlinearity and heavy-tailed distributions. Activities are ranked by $|\rho_i|$ and displayed as a tornado diagram. High-$|\rho_i|$ activities need not lie on the deterministic critical path: iteration exposes *near-critical path* activities—those with low deterministic total float that become critical under specific duration realisations—which is the principal motivation for QSRA over deterministic CPM alone.